A Novel Multiscale Perspective Based Hotspot Temperature Assessment Method for Film Capacitor in DC-Link Applications

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***Abstract*—DC-Link has been widely used in power electronics applications like photovoltaic systems, electric vehicles, aircraft, and LED drivers. Film capacitors often present a thermal management bottleneck in DC-Link. A significant number of novel film materials have been designed to enhance the high-temperature performance of film capacitors. However, the cost of fabricating these materials into capacitor cores and conducting tests is exceedingly high. Therefore, proposing an efficient modeling method to evaluate the hotspot temperature of film capacitors becomes a burning issue. This article proposes a multi-scale finite element modeling (FEM) method to determine the equivalent characteristics of metallized films composed of various materials, as well as the temperature rise in film capacitor units. Initially, equivalent electromagnetic and thermal micro-scale models of multilayer metallized films are established to analyze the principle of film capacitors and obtain equivalent parameters of metallized polymer film. Subsequently, the temperature rise of the capacitor unit is simulated using the macro-scale model, with validation provided by temperature rise tests and data from product manuals. The proposed approach is straightforward, making it applicable to capacitors with various polymer materials and different shapes, and the error in hotspot temperatures obtained through this method is 2.20%.**

***Index Terms*—DC-Link, film capacitor, finite element method, thermal management.**

I. Introduction

In DC-Link applications, capacitors are used to absorb the ripple current generated by converter switches, thereby suppressing voltage fluctuations and ensuring stable and smooth voltage and current on the DC side [1], [2], [3], [4], [5]. Typical configurations of power electronic applications with DC-link capacitors and the simplified lumped model of a film capacitor are illustrated in Fig. 1. Such topologies have been widely used in power electronics applications, including wind turbines, photovoltaic systems, electric vehicles, aircraft, and LED drivers. In this figure, C represents the capacitance value, R_{D} denotes the insulation resistance of the capacitor, R_{ESR} indicates the equivalent series resistance (ESR) of the capacitor, and L_{ESL} refers to the equivalent series inductance of the capacitor [6].

Compared to electrolytic capacitors, film capacitors are not affected by electrolyte evaporation. Additionally, the capacitance

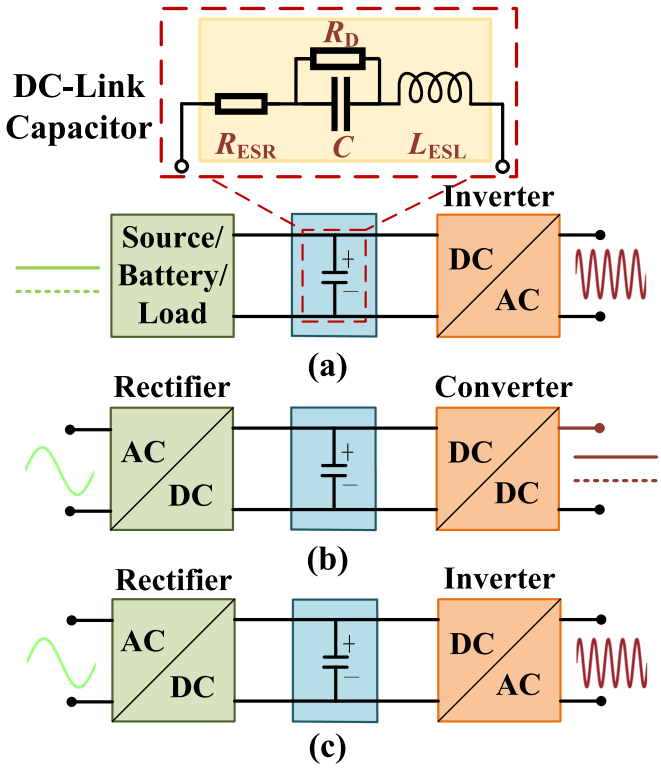


Fig. 1. Typical configurations of power electronic conversion systems with DC-Link capacitors and the simplified lumped model of capacitors: (a) Source/battery/load-DC-AC converters. (b) AC-DC-DC converters. (c) AC-DC-AC converters.

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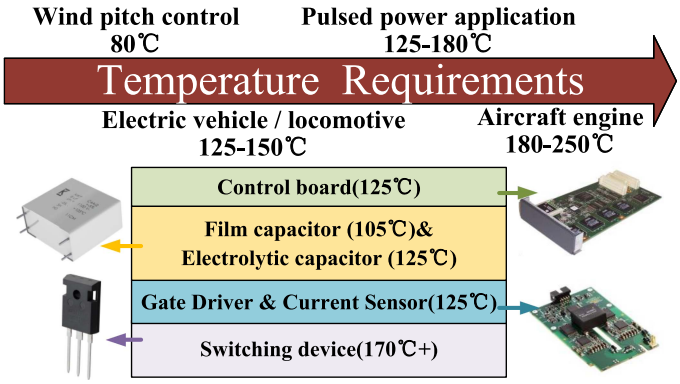


Fig. 2. Typical maximum operating temperatures of components and the high-temperature conditions encountered in some applications.

and ESR of film capacitors remain relatively stable across a wide frequency range and exhibit lower temperature sensitivity [1], [4]. Moreover, film capacitors possess self-healing properties, when a breakdown occurs, the electrode material around the breakdown point will instantly be vaporized by electric arc, allowing the capacitor to continue functioning with some loss in capacitance. To further enhance their durability, segmented metallization technology is employed in film capacitors, this approach can effectively limit the spread of damage by incorporating non-metallic zones on the electrodes, significantly boosting the durability of film capacitors and making them an ideal choice for power electronics applications [7], [8]. However, the maximum operating temperature of film capacitors is lower than that of the switching devices and digital signal processing (DSP) modules, thereby becoming a bottleneck for thermal management of DC-Link applications. Fig. 2 illustrates the typical maximum operating temperatures of components and the high-temperature conditions encountered in some applications [4], [9].

Moreover, with the increasing complexity of power equipment, capacitors are facing with more challenging operating conditions, including exposure to high temperatures and intense electromagnetic interference, at the same time, the increasing demand for miniaturization restricts the heat dissipation area of capacitors, adversely affecting their lifetime. Due to the low price of film capacitors, they are not worth repairing after being damaged. Consequently, the "Mean Time Between Failures" (*MTBF*) of a film capacitor can be seen as the lifetime of the capacitor unit (L). In reliability analysis, the failure rate of capacitors (λ) is commonly expressed using "Failure in Time" (*FIT*), which numerically represents the reciprocal of the capacitor's fault free operating time, so the unit for λ is expressed as (h^{-1}). According to IEC-61709, the failure rate of capacitors is influenced by temperature stress factors and the reference failure rate [10]:

$$\begin{aligned}\lambda &= \lambda_{\mathrm{ref}} \times \pi_{\mathrm{T}} \\ &= \lambda_{\mathrm{ref}} \times \exp\left[\frac{\mathrm{E_a}}{\mathrm{k_0}}\left(\frac{1}{\mathrm{T_{ref}}} - \frac{1}{\mathrm{T_h}}\right)\right] \end{aligned} \tag{1}$$

And the simplified lifetime formula of film capacitors is as follows:

$$\begin{aligned} L = \frac{1}{\lambda} &= \frac{1}{\lambda_{\mathrm{ref}} \times \exp\left[\frac{\mathrm{E_a}}{\mathrm{k_0}}\left(\frac{1}{\mathrm{T_{ref}}} - \frac{1}{\mathrm{T_h}}\right)\right]} \\ &= \mathrm{L_0} \times 2^{\frac{\mathrm{T_{ref}} - \mathrm{T_h}}{10}} \end{aligned} \tag{2}$$

where, π_T is temperature stress factors, λ_{ref} is the reference failure rate (in h^{-1}), E_a is the activation energy, k_0 is the Boltzmann constant, L_0 is the lifetime of film capacitor under rated hotspot temperature (in h), T_h and T_{ref} is the operating hotspot temperature and rated hotspot temperature of capacitor (in °C), respectively.

The common upper working temperature limit for film capacitors is 105 °C, due to the inverse relationship between the lifetime and failure rate of capacitors, when the ambient temperature exceeds the specified operating temperature, the lifetime of capacitors decreases significantly. A typical example is when the hotspot temperature of KEMET C4AQ capacitor increases from 85 °C to 125 °C, the expected lifetime of the capacitor is decreased dramatically from 100,000 hours to 200 hours [11]. Literature [12] points out that the DC-Link capacitors and power modules are the most vulnerable devices in power electronics system, and the failure rate of DC-link capacitors increases much faster than that of power modules during operation. Consequently, DC-Link capacitors have become one of the most risk-prone components in power electronic systems, severely affecting the overall reliability of these systems [13]. Therefore, extensive efforts have been made to enhance the operational capabilities of capacitors at high temperatures by designing new polymer materials. Several researchers have attempted to enhance the high-temperature performance of polymer films by employing composite structures, adding nanofillers and using graft modification. These approaches can impede charge transport within the polymers and suppress high-field conduction under high temperatures [13], [14], [15]. Although some studies have successfully developed polymer materials with excellent high-temperature performance, the complex and costly processes of winding these materials into cores and encapsulating them into capacitor cases have hindered practical testing. Consequently, most of these novel materials have not been used in actual capacitors. Currently, DC-Link capacitor products produced by manufacturers still predominantly use metallized polypropylene films as the capacitor core, as seen in products like KEMET-C4AQ, TDK-B3277, and Faratronic-C4D. [11], [16], [17]. Furthermore, even if these novel materials have been fabricated into capacitor cores, measuring the hotspot temperature of the capacitor requires installing thermocouples inside the capacitor, which would alter the capacitor's structure and may bring some errors into measurement results. Therefore, it is of significance to propose a modeling method that can accurately assess the hotspot temperature rise in film capacitors made from different materials [18]. By obtaining the hotspot temperature of capacitors, the expected lifetime can be calculated, and the replacement schedule can be planned accordingly. Simulations of film capacitors can be divided into two levels: the micro-scale

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and the macro-scale. Micro-scale simulations are utilized to analyze the principles of film capacitors and determine the equivalent characteristics of the cores, while macro-scale simulations are used to assess the hotspot temperature of film capacitors. Based on analysis results, the evaluation of the capacitor's lifetime and unreliability can be conducted.

Existing thermal modeling research on film capacitors typically focuses on the macro-scale level. Common modeling methods include establishing thermal resistance networks of capacitors for rapid determination of thermal distribution in capacitor banks and creating finite element models of film capacitors to analyze their temperature distribution: [19] underscored the stricter volume constraints imposed by high power density in power electronics applications and proposes a rapid method for extracting the thermal resistance of lumped thermal model for capacitor banks. [20] also proposed a similar model and analyzed the function of optimizing parameters for individual capacitors to enhance the lifetime of the capacitor bank. [21] optimized the lumped thermal model of capacitors to account for the effects of ambient temperature changes. [22] established a finite element model for the film capacitor bank and examined how the arrangement and size of capacitor units affect the temperature rise of the capacitor bank. [23] conducted similar research and analyzed the anisotropy of the thermal conductivity of the capacitor core, exploring its effect on the upper limit of internal losses. Similar research have also been conducted on electrolytic capacitors [24], [25]. However, research on micro-scale modeling of capacitor is relatively scarce, some studies have modeled and analyzed the representative internal structures of large film capacitors used in power systems: [26] analyzed film capacitors' operational principles and developed an electrothermal micro-scale model for capacitor windings, using it to assess internal losses in film capacitor banks for modular multilevel converters. [27] proposed a micro-scale modeling method for metallized polymer films in film capacitors and analyzed capacitor characteristics based on this model. Nonetheless, due to variations in capacitor structures and types of materials, these proposed models may not effectively analyze film capacitors used for low power applications. Establishing a corresponding multi-scale model of film capacitors is of great significance for research on the thermal management and lifetime evaluation of film capacitors.

Therefore, this paper primarily addresses the critical issue of evaluating the temperature rise and reliability of film capacitors before manufacturing. It introduces a multi-scale modeling method to assess the hotspot temperature and lifetime of film capacitors, as shown in Fig. 3. The main contributions of this paper are as follows:

1) Proposing a method to determine the equivalent electrical and thermal characteristics of metallized polymer film cores by loss analysis and micro-scale simulation.
2) A macro-scale model of a film capacitor is established by integrating equivalent material parameters obtained from the micro-scale analysis. Electrothermal simulation is conducted to acquire the temperature distribution within the capacitor unit. This approach helps in evaluating the hotspot temperature and assessing the overall reliability of the capacitor under various operating conditions.
3) The results of experiments and calculations indicate that the proposed model achieves sufficient accuracy and maintains simplicity, thereby reducing computation time and enhancing its reusability for film capacitors of various materials and shapes.

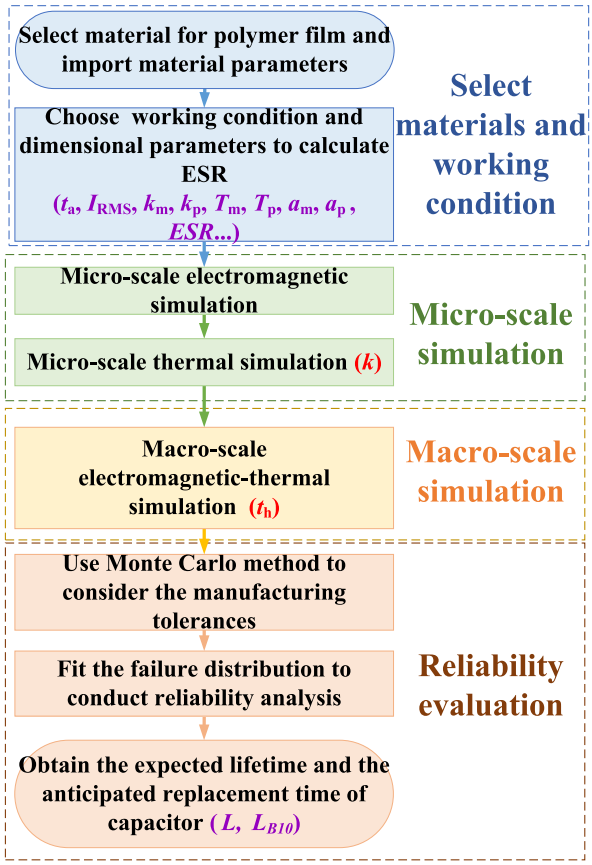


Fig. 3. Multi-scale modeling method for assessing the hotspot temperature and lifetime of film capacitors.

II. Analysis of Film Capacitors for DC-Link

To accurately assess the hotspot temperature of film capacitors, it is essential to analyze the capacitor's structure and loss. This section discusses the types of polymer materials used in capacitor cores, followed by an introduction to the principles and structure of capacitors. After that, an analysis of the internal losses of capacitors is conducted, which includes dielectric loss, electrode loss and leakage loss. These factors are crucial in understanding the thermal behavior of film capacitors in operation.

A. Material and Structure Configuration of Metallized Film Capacitor

The characteristics of film capacitors are significantly influenced by the type of polymer film. Consequently, film capacitors are commonly classified based on the type of dielectric material

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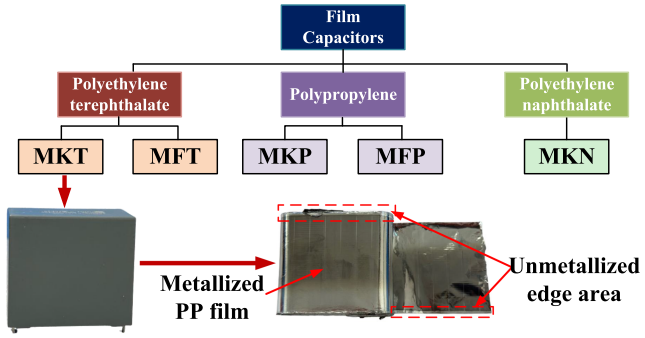


Fig. 4. Classification and identification codes for film capacitors.

TABLE I
COMPARISON OF PARAMETERS OF POLYMER MATERIALS

Material	PP	PET	PEN
Dielectric constant (ε_r)	2.2	3.2	3.0
C temperature coefficient (in 10^{-6}/K)	-250	+600	+200
Dissipation factor (1kHz)	0.0002	0.0050	0.0040
Time constant (in s)	100000	25000	25000

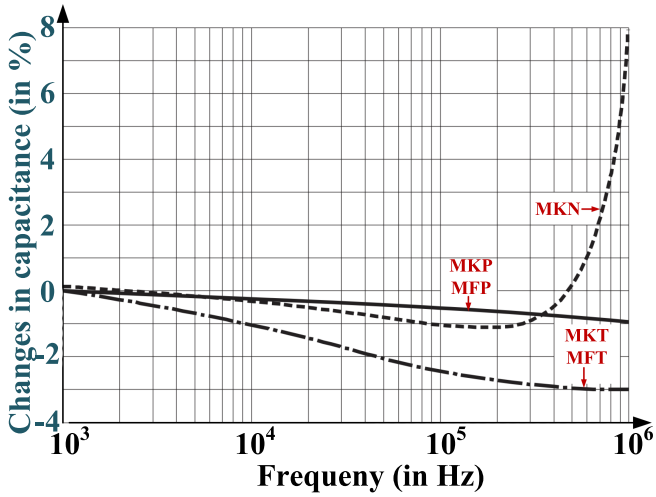


Fig. 5. Variation in capacitance values of film capacitors with different materials across frequencies.

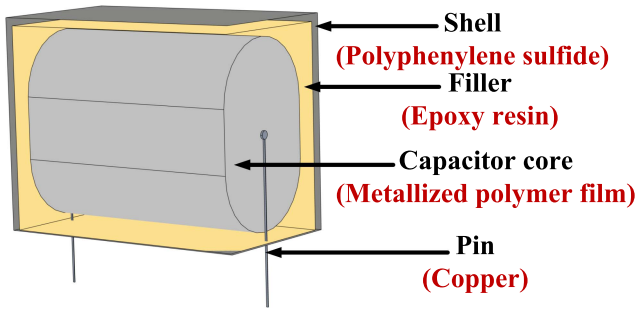


Fig. 6. Internal structure and materials of film capacitors.

used. According to UNE-EN 60062:2005, the classification and identification codes for film capacitors are illustrated in Fig. 4. Among them, "M" stands for metallization, while "K" or "F" on the second character denotes metallized plastic film or metal foil, respectively. The last character of the code indicates the type of dielectric: "T" for polyethylene terephthalate (PET), "P" for polypropylene (PP), and "N" for polyethylene naphthalate (PEN) [28].

Table I shows the principal parameters of polymer materials used in film capacitors [29]. Compared to other materials, polypropylene exhibits lower loss factor, which is beneficial in reducing dielectric loss and extending the lifetime of capacitors in power electronic applications. Furthermore, Fig. 5 illustrates the variation in capacitance values across frequencies for capacitors made from different materials, showing that film capacitors made with polypropylene film exhibit minimal fluctuation across various frequencies. So, the commercially available DC-LINK film capacitors predominantly utilize metalized polypropylene film as the capacitor cores [16], [17], [29]. Therefore, this paper opts to analyze film capacitors made of metallized polypropylene films (MKP).

In typical cases, metallized film capacitors have a cylindrical shape with a polycarbonate core axle at the center. Hundreds of layers of metallized polymer films are wound around this core axle, and the terminals of the capacitor core are sprayed with layers of nano zinc. The metallized polymer film consists of a polymer film about 5 μm thick and an aluminum electrode layer about 20 nm thick serving as the electrode. Adjacent film layers are slightly offset, and the width of the aluminum electrode layer is smaller than that of the polypropylene film to ensure that each metallized electrode is connected only to one terminal of the core [5]. In DC-Link applications, flattened film capacitors are widely used. During the manufacturing process, the core axle is removed, and the cylindrical core is flattened. This operation can increase the volume efficiency and eliminates air between the film layers, reducing capacitance loss caused by air breakdown during operation. The capacitor core is placed into a polyphenylene sulfide (PPS) case and filled with epoxy resin to isolate it from the air. The typical structure of the film capacitor used in DC-Link applications is illustrated in Fig. 6.

As mentioned above, the dimensions of components in film capacitors vary significantly: the capacitor physical size is measured in millimeters, the single-layer polymer film thickness in micrometers, and the metallized electrode thickness in nanometers. Additionally, one capacitor core consists of hundreds of wound film layers, which pose challenges for simulation setup. To address this, the hundreds of layers of metallized polymer film are merged and equivalently represented as an anisotropic material. The essential parameters required for this characterization include the ESR and the thermal conductivities in axial and radial directions [30].

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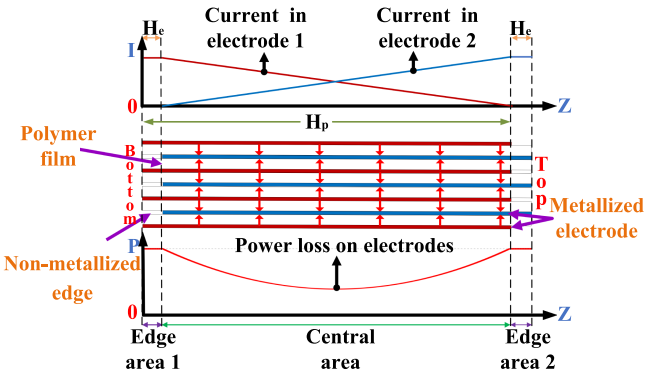


Fig. 7. Distribution of current and electrode loss on various parts of the film capacitor core.

B. Loss Analysis of Metallized Film Capacitor

The power loss within the capacitor core consists of three components: electrode loss generated by the capacitor current on the metallized electrodes, dielectric loss generated by the polymer film in alternating electric field, and leakage loss generated by conduction current through the dielectric.

Dielectric loss is uniformly distributed within the polymer film, and its value is correlated with the reactive power and dissipation factor (tan(δ)) of the polymer material:

$$P_d = Q\tan(\delta) = \omega C U^2 \tan(\delta) = \frac{I_{rms}^2 \cdot \tan(\delta)}{2\pi f} \times \frac{T_P}{\varepsilon_0 \varepsilon_p D \times (H_p - H_e)} \tag{3}$$

$$D = \frac{S}{T_m + T_p} \tag{4}$$

where, P_d is the dielectric loss of the capacitor (in W), Q is the reactive power of the capacitor (in Var), S is the cross-sectional area of the core (in m^2), T_m and T_p is the thickness of electrode and polymer film (in m), respectively. H_p is the height of the metal-coated section on polymer film (in m), and H_e is the height of non-metallized edge on one side (in m), D is the total length of metallized polymer film (in m). It should be noted that the tan(δ) of same polymer material will change across different frequencies, and the total dielectric loss of a capacitor is the sum of dielectric loss under various frequencies.

The electrode loss is also distributed within the capacitor core, but this distribution is non-uniform due to the electrodes' lower height compared to the polymer film and the offset alignment between the two layers of metallized polymer films. For analyzing electrode loss, the capacitor core can be divided into two edge regions and a central region, as illustrated in Fig. 7. It can be observed that the current distribution on the two electrodes is symmetric: In the central region, current is present on both electrodes with a linear variation in current values. In the edge regions, current exists only on one electrode and remains constant throughout this region [22].

When the height Z is less than H_e, the current value on the electrode connected to the bottom remains at its maximum. When Z exceeds H_e but is still less than H_p, the current in this electrode linearly decreases, reaching zero when Z equals H_p. The loss distribution on the electrode connected to the bottom terminal of the core can be expressed by (5):

$$dp_{e_b} = \begin{cases} I^2 \left(\rho_m \frac{dz}{\frac{D}{2}\cdot T_m}\right) & (0 \le Z \le H_e) \\ I^2 \frac{(H_p - Z)^2}{(H_p - H_e)^2} \left(\rho_m \frac{dz}{\frac{D}{2}\cdot T_m}\right) & (H_e \le Z \le H_p) \end{cases} \tag{5}$$

where, dp_{e_b} is the linear density of the loss on the electrode connect with bottom terminal, ρ_m represents the resistivity of the electrode material. Due to the symmetry of current and loss distribution in the two electrodes, the total electrode loss (P_e, in W) in film capacitor can be calculated as:

$$P_e = \frac{4}{3} I^2 \left(\rho_m \frac{H_p + 2H_e}{D T_m}\right) \tag{6}$$

The voltage applied across the DC-Link capacitor is a DC voltage with ripple. Under DC voltage, some leakage current will flow through the dielectric of the capacitor and generate leakage loss. According to IEC 60384-1:2016, the insulation resistance of the capacitor is used to describe the DC resistivity of the capacitor [31]. This resistance is independent of frequency and only related to temperature. When temperature changes, the measured values need to be multiplied by corresponding correction factors. The insulation resistance data provided by manufacturers includes resistance values and time constants (τ), with the conversion relationship given by $\tau = \mathrm{R_D} \times \mathrm{C_R}$ (in M$\Omega \cdot \mu$F). In fact, the insulation resistance of existing DC-Link capacitor products is extremely high (e.g., KEMET-C4AQ: 300 MΩ; Faratronic-C3D: $\geq$10,000 s or $\geq$90.91 MΩ). Considering that the common DC voltage limit for such capacitors is about 500 V, the actual leakage current is thus very small, and the leakage loss remains very low (e.g., 100 μF KEMET-C4AQ: $\leq$0.833 mW; 110 μF Faratronic-C3D: $\leq$2.75mW) [11], [16], [17]. The value of leakage loss exhibits a difference of two to three orders of magnitude compared to the overall internal losses of the capacitor. Considering that the leakage loss value is extremely small and difficult to calculate before capacitor manufacturing, it is acceptable to ignore the leakage loss during loss and ESR calculation.

Based on the above analysis, the internal losses, including dielectric loss and electrode loss, can be expressed by the loss on the ESR in the equivalent circuit of the film capacitor shown in Fig. 1, in low frequency application, the ESR is primarily determined by dielectric loss, which decreases with the increase in frequency. In medium to high frequency applications, electrode loss constitutes the major part of internal losses. The total internal losses (P) and ESR of the film capacitor are given by (7) and (8). The total internal loss is the sum of loss of different frequency components:

$$P = \sum_{\mathrm{i}=1}^{n} R_{\mathrm{ESR,i}} \times I_{\mathrm{rms,i}}^2 \tag{7}$$

$$R_{\mathrm{ESR,i}} = \left[\frac{\tan(\delta) T_P}{2\pi f \varepsilon_0 \varepsilon_p D \times (H_p - H_e)} + \frac{4}{3}\left(\rho_m \frac{H_p + 2H_e}{D T_m}\right)\right] \tag{8}$$

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TABLE II
PARAMETERS OF MEATLLIZED POLYMER FILM

Parameters	Values
Material	polypropylene (film); aluminum (electrode)
Height (in mm)	51.5 (metal-coated section of film); 1 (edge)
Thickness (in mm)	5×10^{-3} (film); 20×10^{-6} (electrode)
Conductivity (in S/m)	3.0×10^{-15} (film); 3.5×10^{7} (electrode)
Thermal conductivity (in W/(m·K))	0.22 (film); 237 (electrode)
Density (in kg/m³)	0.9×10^{3} (film); 2.7×10^{3} (electrode)
Specific heat capacity (in J/(kg·°C))	1.9×10^{3} (film); 0.9×10^{3} (electrode)

where, $R_{\mathrm{ESR,i}}$ is the ESR of film capacitor when the current frequency is i, and $I_{\mathrm{rms,i}}$ is the RMS value of the current component with frequency i.

III. MICRO-SCALE MODELING

Given that a single capacitor's core is wound by hundreds of layers of metallized film, it is impractical to model the internal structure of the capacitor in macro-scale simulations. Therefore, it is necessary to elucidate the principles of film capacitors through micro-scale simulations of representative structures. These simulation results can intuitively demonstrate the working principle of film capacitors and the distribution of potential, while also acquiring equivalent material parameters for metallized polymer films.

A. Micro Scale Electromagnetic Modeling of Multi-layer Metallized Film Winding Structure

Selecting COMSOL Multiphysics as the FEM software, a 10-layer metallized polymer film structure is simulated to obtain the distribution of potential and current in film capacitor.

In the simulation process, the internal distribution of electric potential and fields within the multi-layer film winding structure can be determined by applying the parameters of the dielectric material to the polymer film and setting voltage differences on the corresponding boundaries. Based on the simulation results, the working principles of the film capacitor can be observed. The parameters of the film used in the micro-scale modeling are presented in Table II.

Since the height of polymer films is usually thousands of times larger than their thickness, it is necessary to adjust the display scale during simulation. In this section, the simulated images are stretched 500 times along the direction of film stacking to ensure visual clarity. To maintain meshing and solving efficiency, the electrodes of the metallized layer are positioned at the boundaries of the polymer film, and their thickness is provided as a parameter when setting the electrical shielding boundary conditions and thin layer heat transfer conditions. Fig. 8 depicts the boundary condition of simulation of the film. The blue and red arrows in Fig. 8(a) represent the boundary conditions for

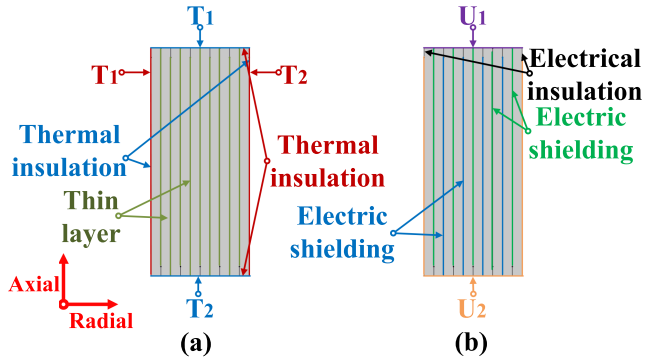


Fig. 8. The boundary condition of micro-scale simulation, (a) Thermal simulation, (b) Electromagnetic simulation.

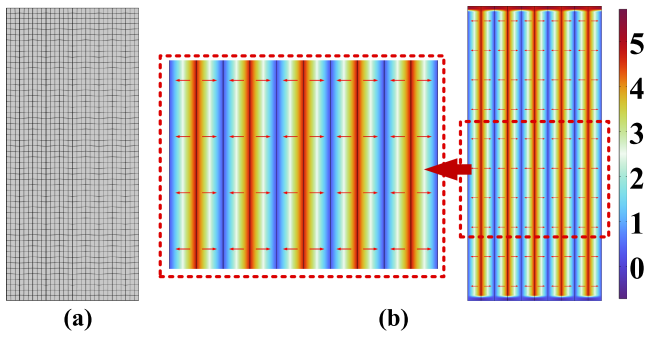


Fig. 9. (a) Mesh division of micro-scale 2D simulation of 10 layers of metallized polymer films, (b) Distribution of potential and the direction of the electric field (in V).

axial and radial thermal field simulations, respectively. While the purple and yellow arrows at locations U_1 and U_2 in Fig. 8(b) indicate the boundary conditions for the electromagnetic simulation. The simulation is carried out in the frequency domain, with a simulation frequency of 50 Hz [1].

Fig. 9(a) shows the mesh division for micro-scale simulation, consisting of 1590 quadrilateral elements. In theory, by applying a specific potential difference between the two edges along the axial direction and setting the boundaries in another direction as electrically insulating boundaries, the distribution of current and potential inside the stacked films can be obtained. This setting is reasonable when simulating a complete capacitor winding. However, there is a potential issue that in a film capacitor, since only the edge closest to the center is not in contact with the metallized electrode, using the above boundary conditions might lead to a lack of electric field in certain regions, thereby inaccurately reflecting the electric field distribution in the capacitor. To address this problem, it is necessary to retain the metallized electrode on the next layer of film at the boundary and assign the corresponding potential on it during the simulation process.

Fig. 9(b) shows the distribution of potential in film capacitor, the top terminal and its connected electrodes are maintained at a high potential, while the bottom terminal of the film and its connected electrodes are maintained at a low potential. Additionally, the red arrows in the figure illustrate the direction of the electric

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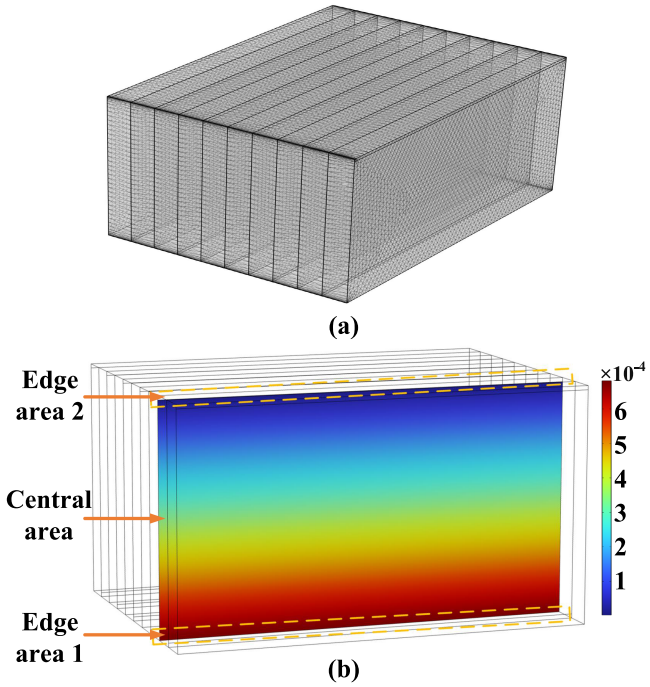


Fig. 10. (a) Mesh division of micro-scale 3D simulation of 10 layers of metallized polymer films, (b) Current density modulus on electrode connect with bottom terminal (in A/m).

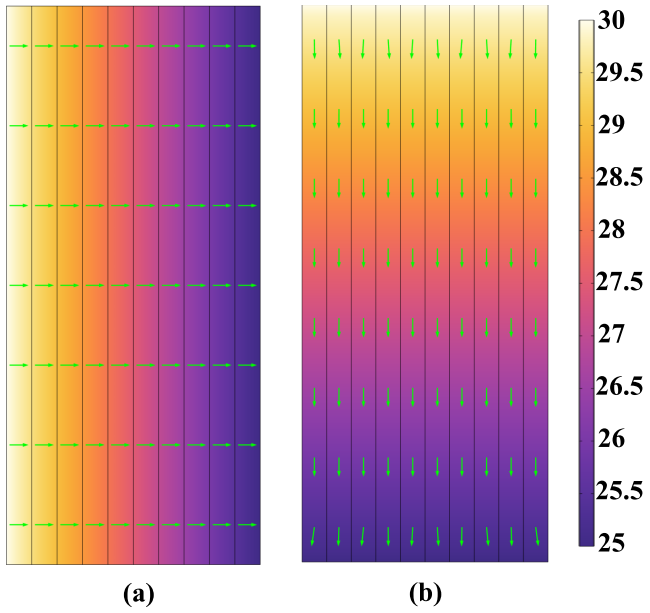


Fig. 11. Temperature distribution of 10 layers metallized polymer films stacking structure after applying 5 °C temperature difference (in °C) (a) Radial thermal simulation, (b) Axial thermal simulation.

field within the structure, pointing from the high-potential electrode plates to the adjacent low-potential electrode plates, with the polymer film serving as the dielectric between these plates. This microstructure forms several parallel plate capacitors that are connected in parallel [32], [33].

The two-dimensional model is stretched to construct a three-dimensional micro-scale model. The frequency domain simulation was also performed on the three-dimensional microstructure of multilayer metallized polymer films to obtain the current distribution on the electrodes. The mesh division of the three-dimensional model is shown in the Fig. 10(a), consisting of 129020 tetrahedral elements. The simulation results of the current distribution on the electrode connected to the bottom terminal are displayed in Fig. 10(b). The current remains maximal in the bottom-edge area and decreases linearly in the central area, reaching zero at the edge of another edge area [34], [35].

B. Micro Scale Thermal Modeling of Multi-layer Metallized Film Winding Structure

Considering the anisotropy of the capacitor core's thermal conductivity, it is essential to determine its thermal conductivity in different directions before conducting macro-scale simulations. The equivalent thermal conductivity of the metallized film core in two directions can be determined through micro-scale FEM. To calculate the thermal conductivities in different directions, the capacitor core must undergo two simulations within the same coordinate system, each with distinct thermal boundary conditions. When calculating thermal conductivity in one direction, the influence from the other direction must be eliminated. Therefore, when determining radial thermal conductivity, 5 °C temperature difference is set on the two edges in the radial direction, while the axial edges are designated as thermal insulation boundaries, and vice versa. The boundary conditions for micro-scale thermal simulation are depicted in Fig. 8(a). Furthermore, the effective thermal conductivity of the metallized film core can be calculated based on the conductivity and thickness proportions of the materials [22], [36]. The thermal conductivities (k, in W/(m·K)) in the two directions can be expressed as follows:

$$q = -k\nabla T \tag{9}$$

$$q_{avg} = \frac{1}{A_{\text{sec}}} \int q ds \tag{10}$$

$$\nabla T = \frac{T_1 - T_2}{W} \tag{11}$$

$$k = \frac{q_{avg} \times W}{T_1 - T_2} = \begin{Bmatrix} \frac{k_p \cdot k_m}{k_p \cdot \alpha_m + k_m \cdot \alpha_p} & \text{(radial)} \\ k_p \cdot \alpha_p + k_m \cdot \alpha_m & \text{(axial)} \end{Bmatrix} \tag{12}$$

where q_{avg} is the average heat flux (in W/m^2), q is the heat flux in the direction being studied, T_1 and T_2 represent the temperatures at the boundaries of the film (in °C), A_{sec} is the cross-sectional area (in m^2), α_p and α_m represent the thickness ratios of the polymer film and the electrode to the total thickness, respectively, while k_p and k_m is the thermal conductivity of polymer film and metal electrode (in W/(m·K)), respectively. W is the distance between the two boundaries where the temperature difference is applied (in m).

Fig. 11 shows the temperature distribution within the 10-layer metallized polymer films after a 5 °C temperature difference is

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applied to the different boundaries of the film. In radial direction, the polymer film and metallized electrodes are arranged in series, while in axial direction, they are configured in parallel. Owing to the substantial thickness difference between the polymer film and the electrodes, the thermal conductivity on radial direction is almost equivalent to that of the polymer film in series arrangement, resulting in a uniform distribution of the radial heat flux as shown in Fig. 11(a). While the thermal conductivity in the axial direction shows a considerable difference compared to any homogeneous material, this is due to the parallel arrangement of the polymer films and metallized electrodes. Additionally, because the electrode thickness at the edge region is only half that in the central area, there is a significant decrease in thermal conductivity at the edge compared to the center area. Based on the parameters of the metallized polymer film are listed in the Table II, the radial thermal conductivity is 0.2209, while the axial thermal conductivities of the central and edge regions are 1.1633 and 0.6926, respectively, resulting in a distortion of the thermal flux direction as shown in Fig. 11(b).

IV. Macro-Scale Modeling and Experiment

In contrast to micro-scale simulations, which are used for fundamental analysis, macro-scale simulations are utilized to ascertain the temperature of hotspots in film capacitors. Due to the relatively complex internal structure of the core, the capacitor core is treated as a resistor in macro-scale simulations. By setting the corresponding resistivity on the capacitor core, the resistance of the capacitor core is consistent with the calculation result of (8), so that the loss values on the core are consistent with the calculated result of (7), the resistivity of core (ρ_{c_e}, in $\Omega \cdot$ m) is as follows:

$$\rho_{c_e} = \frac{ESR \cdot S}{H_p + H_e} \tag{13}$$

A. Macro-Scale Electrothermal Simulation of Capacitor Unit

In addition to the equivalent parameters obtained in the previous sections, the macro-scale simulation also requires the thermal capacitor and mass of the core. The density (ϱ_e, in kg/m^3;) and specific heat capacity (C_{th_e}, in J/(kg·°C)) of the metallized polymer film can be determined by (14) and (15), respectively:

$$\varrho_e = \varrho_p \cdot \alpha_p + \varrho_m \cdot \alpha_m \tag{14}$$

$$C_{th_e} = \frac{C_{th_p} \cdot \varrho_p \cdot \alpha_p + C_{th_m} \cdot \varrho_m \cdot \alpha_m}{\varrho_p \cdot \alpha_p + \varrho_m \cdot \alpha_m} \tag{15}$$

where, ϱ_e, ϱ_p and ϱ_m are the density of the metallized polymer film, polymer film and metal electrode, C_{th_p} and C_{th_m} are specific heat capacity of polymer film and metal. And the hotspot temperature of film capacitor ($T_{\rm h}$, in °C) can be calculated through thermal resistance between the ambient and hotspot (R_{thh-a}, in °C/W), internal losses (P, in W), and ambient temperature ($T_{\rm a}$, in °C):

$$T_h = P \times (R_{thh-a}) + T_a \tag{16}$$

TABLE III
Parameters of Capacitor Unit

Parameters	Values
Capacitance (in μF)	110
Capacitance Tolerance (in %)	±10
Expected lifetime (in h)	100 000h (T_h=70℃)
Polymer materials	polypropylene
Packaging materials	polyphenylene sulfide (case), resin (filler)
tan(δ)	2.0×10^{-4}
Size (in mm)	57 (width), 50 (height), 35 (thickness), 52.5(width of core)
dV/dt (in V/μs)	15
$R_{thh\text{-}a}$ (in °C/W)	10
S (in mm^2)	1156.2
D （in m）	230.3187

TABLE IV
Parameters of Materials for Simulation

Parameters	Values
Thermal conductivity (in W/(m·K))	1.1362 (metallized polymer film-axial); 0.2209 (metallized polymer film-radial); 0.15 (resin); 15 (PPS); 400(copper)
Density (in kg/m^3)	907.17 (metallized polymer film); 1820 (resin); 1650 (PPS); 8960(copper)
Specific Heat Capacity (in J/(kg·°C))	1888.14 (metallized polymer film); 1340 (resin); 1000 (PPS); 385(copper)

A model of the 110μF Faratronic C3D film capacitor was established in COMSOL Multiphysics. The thermal conductivities and other parameters obtained from previous sections were applied to the core, and a current of 7.36 A was injected into the terminals of the capacitor core, the frequency was set at 50 Hz, the ambient temperature was maintained at 30 °C. The parameters used for the capacitor in the simulation are presented in Table III, and the material parameters are listed in Table IV [17].

Based on (8), the ESR of the capacitor was calculated at frequencies of 50 Hz and 100 Hz, with the calculation parameters shown in Tables II and III. The ESR of the capacitor was also measured using a Tonghui TH2832 LCR meter; the calculated ESR values were 0.01449 Ω at 50 Hz and 0.00746 Ω at 100 Hz. The experimental results were 0.01417 Ω and 0.00735 Ω, respectively, resulting in discrepancies of 2.26% and 1.50%. Based

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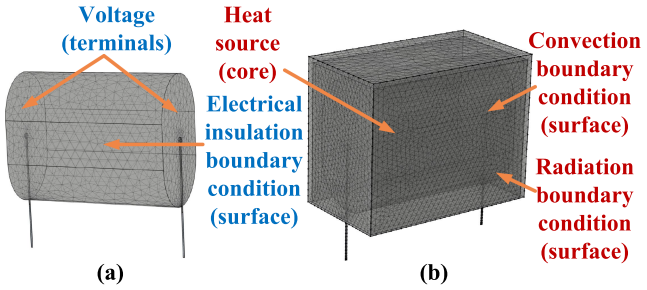


Fig. 12. Boundary condition of macro-scale simulation (a) Electromagnetic simulation, (b) Thermal simulation.

on the calculation results at 50 Hz and the core dimension parameters listed in Table III, the equivalent resistivity of the core was determined to ensure consistency between the core's resistance in FEM and the calculated result.

Fig. 12(a) and (b) show the boundary conditions for macro-scale simulations; the mesh consists of 57572 tetrahedral elements. To improve the calculate speed, the simulation was divided into two steps: electromagnetic simulation was performed in the frequency domain, followed by steady-state thermal simulations to analyze the temperature rise of the capacitor based on the results of the electromagnetic simulations. Capacitors dissipate heat through thermal radiation and natural convection. According to the data provided by [25], the heat transfer coefficient of capacitors is 9.49, and the surface emissivity is 0.85. Due to the structure and application environment of the film capacitor is complex, some simplifications were made in both the simulation and experimental setups [37], [38], [39], [40]:

1) The thickness of the edge area (1 mm on each side) is significantly smaller than that of the central area (50.5 mm). Therefore, the two edge areas are not modeled separately in the geometric representation. Instead, the overall equivalent thermal conductivity in the radial direction of the capacitor core is calculated based on the series relationship, then set on the core.
2) In the actual structure of the film capacitor, several layers of non-metallized polymer film are wound around the outer side of the core to prevent the stacked films from loosening and to ensure insulation on the side of the core. In order to improve computational efficiency, they were not explicitly modeled geometrically in the simulation. On the contrary, the corresponding settings are made on the boundaries in contact with the core.
3) It is assumed that the power loss in the model is only distributed in the capacitor core and the metal coating layer. Considering the relatively small resistance of the pins, the impact of loss on the pins on the temperature rise of the capacitor can be neglected [25].
4) The voltage across the two terminals of a capacitor is composed of a DC voltage with an AC voltage ripple. Since extensive research has demonstrated that the DC voltage component has almost no impact on the thermal effects of the capacitor, an AC voltage was directly applied on the capacitor terminals during the experimental process [41].

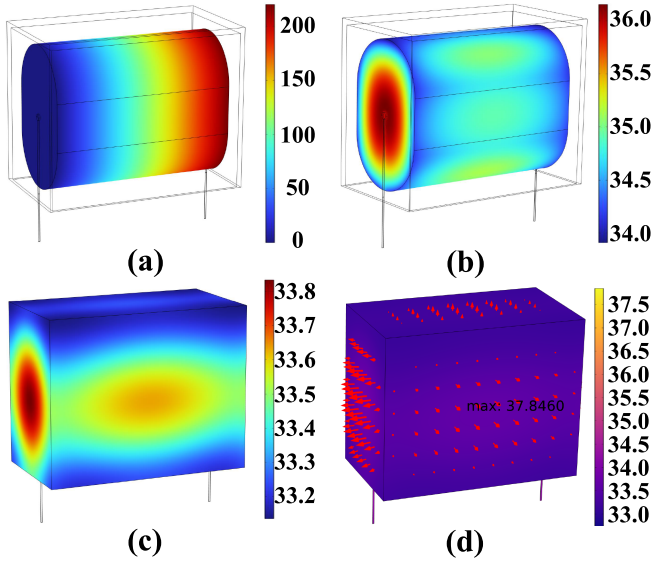


Fig. 13. (a)The distribution of electric potential on capacitor core (in V), (b) The distribution of temperature on capacitor core surface (in °C), (c) The distribution of temperature on capacitor case (in °C), (d)The distribution of heat flux of capacitor (in °C).

Fig. 13 shows the result of macro-scale FEA modeling of the capacitor, Fig. 13(a) and (b) show the potential and temperature distributions on the capacitor core, respectively. In Fig. 13(a), the hundreds of layers of metallized films composing the capacitor core has been considered as a whole, the simulated electric field is uniformly distributed between the terminals of the core, which differs significantly from the actual situation, emphasizing the importance of conducting micro-scale electromagnetic simulations. Fig. 13(b) illustrates the temperature distribution on the surface of the capacitor core, it can be observed that the surface temperature along the axial direction is higher than that along the radial direction. This is due to the axial thermal conductivity of the core being much greater than the radial thermal conductivity, with most of the heat generated in the core conducted through the axial direction. Fig. 13(c) illustrates the temperature distribution on the capacitor surface, the maximum and minimum temperature on the capacitor surface is 33.83 °C and 33.14 °C, respectively. The red arrows in Fig. 13(d) indicate the distribution of heat flux; the heat flux along the axial direction is more densely distributed compared to other directions. The hotspot of the capacitor unit is located in the middle of the core, reaching 37.8460 °C.

B. Experimental Verification of Capacitor Temperature Rise

Experiments were conducted to validate the simulated results of capacitor surface temperature. Tonghui TH2832 LCR meter was used to measure the capacitance and ESR, while an AC power source was used to drive the capacitor. The experiment, which lasted for four hours, was sufficient to stabilize the capacitor's temperature. Fig. 14(a) illustrates the equipment used

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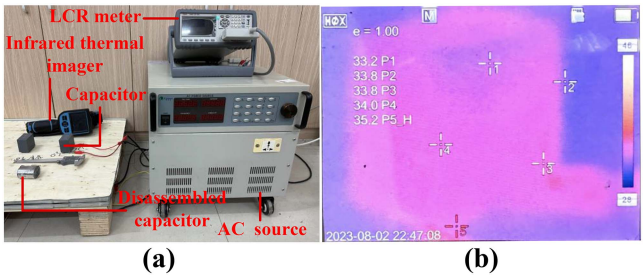


Fig. 14. (a) Experiment equipment, (b) Experiment results of capacitor case temperature rise (in °C).

TABLE V
EXPERIMENT AND SIMULATION RESULTS

Parameters	Values
Capacitance (μF)	110.0 (nominal value), 105.521 (experiment)
ESR (Ω)	0.01449 (calculation), 0.01417 (experiment)
Internal Losses (W)	0.7849 (calculation), 0.7676 (experiment)
Point 1 temperature (°C)	33.2 (simulation), 33.2 (experiment)
Point 2 temperature (°C)	33.8 (simulation), 33.5 (experiment)
Point 3 temperature (°C)	33.8 (simulation), 33.4 (experiment)
Point 4 temperature (°C)	34.0 (simulation), 33.6 (experiment)
Hotspot temperature (°C)	37.8460 (simulation), 37.6770 (calculation)

in the experiment. Due to that it was not feasible to modify the capacitor structure in the laboratory for direct measurement of the capacitor's hotspot temperature, the ambient temperature, internal losses measured in experiment and the thermal resistance parameters from the product manual were used to calculate the actual hotspot temperature of the capacitor. Fig. 14(b) shows the surface temperature rise of the capacitor obtained using an infrared camera, where points P1-P4 are located on the exterior of the capacitor case and point P5 is the temperature of the clamp connected to the pin. The calculations, simulations, and experimental results were compared in Table V. Among them, the internal losses and ESR are calculated using (7) and (8), respectively, while the temperatures are the results of macro-simulation. The error of the hotspot temperature is 2.20%.

Considering that the internal loss of the capacitor is not constant under different conditions, it is necessary to analyze the temperature rise at the hotspot under various ambient temperatures and losses to further evaluate the effectiveness of the proposed modeling method. Additionally, the results were compared with calculation results based on the data provided by the manufacturer. The hotspot temperatures obtained through simulation and calculation under different ambient temperatures and loss are compared in Fig. 15, it can be seen that the error increases with the rise in internal losses, but the overall error still remains within a small range, with the maximum error less than 4.33%.

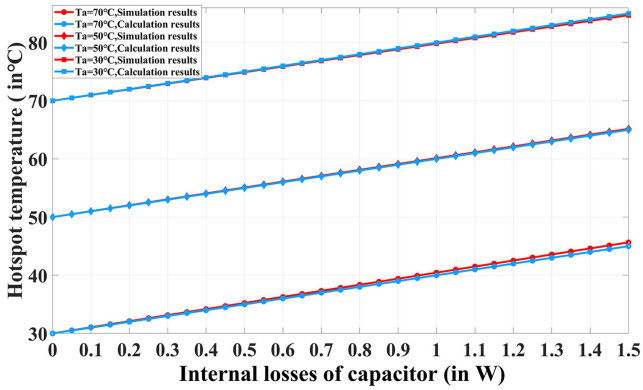


Fig. 15. Comparison of capacitor hotspot temperatures obtained through simulation and calculation under different ambient temperatures and losses.

C. Reliability Assessment of Capacitor

Considering the purpose of obtaining the hotspot temperature of film capacitor is to predict the lifetime and to schedule the replacement time through reliability analysis. Since capacitors are often used in high-temperature environments in practical applications, it is necessary to select a higher ambient temperature for reliability evaluation. Using the proposed method, the temperature rise of a 110 μF C3D capacitor at an ambient temperature of 70 °C is assessed, with the capacitor's internal losses set at 1W. FEM results show that the hotspot temperature reaches 79.8211 °C. Based on (2) and the lifetime data provided in the Table III, the expected lifetime of this capacitor under this working condition is 50623.88 h.

Due to manufacturing tolerances, variations in loss and hotspot temperatures among individual capacitors will lead to differences in the lifetime of capacitors under the same operating conditions. Thus, the uncertainty in device parameters must be considered when assessing the reliability of capacitors. Employing the Monte Carlo method, a Gaussian distribution with a 10% parameter variation is used to simulate differences in capacitor hotspot temperature rise and rated lifetime. MATLAB is utilized to generate 10000 different samples to obtain the distribution of the capacitors' lifetime [7].

Considering that the potential damages resulting from capacitor failure may exceed the costs of replacing them, it is imperative to ascertain the reliability function of capacitors and replace them before their unreliability reaches a critical threshold. B10 life is a common reliability metric used in practice, indicates the time required for 10% of devices to fail under specified operating conditions. The failure distribution of capacitors is modeled using a two-parameter Weibull distribution, with the probability density function $f(t)$ defined as follows [42]:

$$f(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1} e^{-\left(\frac{t}{\eta}\right)^{\beta}} \tag{17}$$

where, η and β represent the scale parameter and shape parameter, respectively, while t denotes the device operating time (in h). And the unreliability function $F(x)$ of the capacitors can be calculated by integrating the probability density function over

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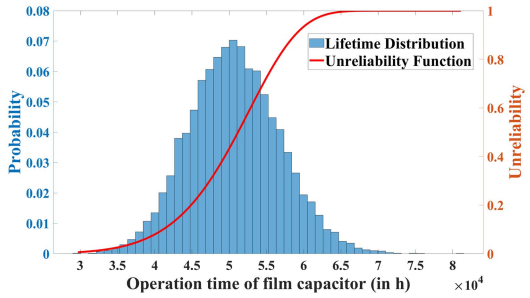


Fig. 16. The distribution of the lifetime of film capacitors and the change in their unreliability over operation time.

time:

$$F(x) = \int_0^x f(t)\,dt \tag{18}$$

The unreliability function represents the probability that a device will fail over a given time, and the function derived from (18) is used to estimate the probability of aging-related failures in capacitors.

As depicted in Fig. 16, the B10 life of the capacitor under this condition is 41164.44 h, which indicates that approximately 10% of the capacitors will fail after operating for 41164.44 hours.

V. Conclusion

This article proposes a multi-scale FEM approach to predict the hotspot temperature rise in film capacitors used in DC-Link applications. A micro-scale FEM model was developed to analyze the electric potential distribution within the capacitor and obtain the equivalent parameters of multi-layer wound metallized polymer films. Subsequently, the temperature rise in the capacitor unit was simulated using a macro-scale FEM model, and the unreliability and lifetime of the capacitors were evaluated based on the simulation results. This method simplifies the model's boundary conditions to maintain simplicity, and its simulation outcomes have been validated through experiments and calculations, the FEM results are close to the experimental results and the data provided by the manufacturer, with a relative error of 2.20% in hotspot temperature. This work could be useful for evaluating the temperature rise and lifetime of film capacitors made from novel film materials, thereby reducing testing expenses. Future directions of this work include two aspects: considering the distribution of loss within the core to achieve more accurate predictions of capacitor hotspot temperatures and analyzing the accelerated aging effects of voltage. Integrating these analyses into the research enables a more accurate lifetime assessment of film capacitors.

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